

INMO(14th) – 1999

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be an acute angled triangle in which D, E, F are points on BC, CA, AB respectively such that AD is perpendicular to BC: AE = EC, and CF bisects $\angle C$ internally. Suppose CF meets AD and DE in M and N respectively. If FM = 2, MN = 1, NC = 3, find the perimeter of the triangle ABC.
 2. In a village 1998 persons volunteered to clean up, for a fair, a rectangular field with integral sides and perimeter equal to 3996 feet. For this purpose the field was divided into 1998 equal parts. If each part had an integer area (measures in sq ft), find the length and breadth of the field.
 3. Show that there do not exist polynomials $p(x)$ and $q(x)$ each having integer coefficients and of degree greater than or equal to 1 such that $p(x)q(x) = x^5 + 2x + 1$.
 4. Let Γ and Γ' be two concentric circles. Let ABC and $A'B'C'$ be any two equilateral triangles inscribed in Γ and Γ' respectively. If P and P' are any two points on Γ and Γ' respectively. Show that $P'A^2 + P'B^2 + P'C^2 = A'P^2 + B'P^2 + C'P^2$.
 5. Given any four distinct real numbers, show that one can choose three numbers say A, b, C from among them such that all the three quadratic equations: $Bx^2 + x + C = 0$, $Cx^2 + x + A = 0$ and $Ax^2 + x + B = 0$.
 6. For which positive integer value of n can the set $\{1, 2, 3, 4, \dots, 4n\}$ be split into n disjoint 4 elements subsets $\{a, b, c, d\}$ such that in each of these sets $a = \frac{b+c+d}{3}$?

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